BubbleGS: Physics-Aware Layers on MCMC Gaussian Splatting for Soap Bubble Reconstruction

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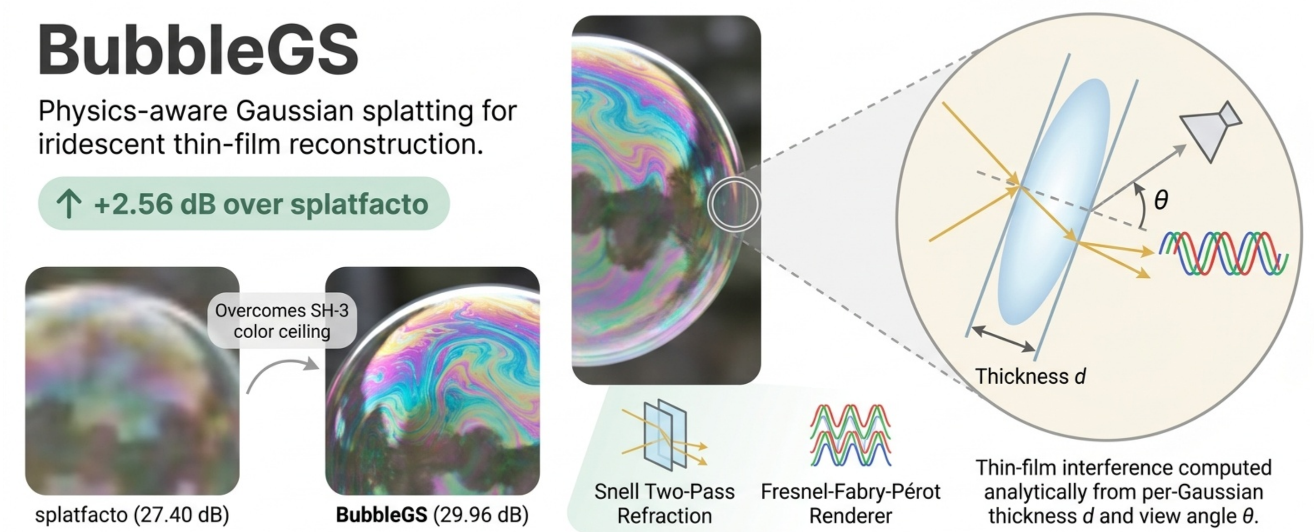


Figure 1. BubbleGS recovers iridescent thin-film appearance. Left: vanilla splatfacto produces a color-flat, milky reconstruction of a soap bubble. Middle: the published splatfacto-mcmc chassis [19] reconstructs the bubble at 29.96 dB across four seeds; this chassis-only gain is prior art. Right: BubbleGS adds composable thin-film physics priors on top of this chassis and improves the LPIPS Pareto frontier at PSNR statistically tied with the chassis.

Abstract

Soap bubbles stress every assumption made by 3D Gaussian Splatting (3DGS): the surface is transparent, the material is a thin film, and its color is set by wavelength-dependent interference rather than diffuse albedo. Vanilla splatfacto fits a captured bubble scene at low PSNR and plateaus well below the data, because low-order spherical harmonics cannot represent the high-frequency Fabry–Pérot oscillation that produces iridescence. We show that splatfacto-mcmc [19] is the right chassis for transparent thin-film 3DGS reconstruction: on `bubblefinal` *this published chassis attains* 29.96 *dB across multiple seeds, a gain we credit to prior art. On top of this chassis we introduce BubbleGS, a composable thin-film physics module: a Snell two-pass refraction layer adapted from Wang et al. [35] and a wavelength-resolved Fabry–Pérot reflectance term adapted from Belcour and Barla [2]. Our best composition advances the LPIPS Pareto frontier; the PSNR delta over the chassis is statistically tied within the measured cross-seed standard deviation. We release reusable nerfstudio-compatible physics layers together with a candid catalogue of negative experiments, and our contribution is the demonstration that thin-film optical priors compose with stochastic Langevin densification without harming convergence.*

1. Introduction

3D Gaussian Splatting [18] reconstructs photorealistic radiance fields by fitting an explicit cloud of anisotropic Gaussians to multi-view images. The representation has matured rapidly through better densification [19, 41], geometryaware variants [8, 10], and material-aware extensions [16, 23, 39]. Yet one class of scene remains an open problem: *transparent thin-film objects* such as soap bubbles, oil slicks, and anti-reflective coatings. Their appearance is set not by diffuse reflection but by wavelength-resolved interference, and the geometry being reconstructed is two surfaces of a single membrane only nanometres apart.

The bubble as a stress test. A soap bubble exposes three failure modes of standard 3DGS in a single scene. First, the membrane is transparent: every camera ray traverses both an outer and inner film boundary, so the deferred-shading assumption used by reflective extensions [39, 45] no longer holds. Second, the perceived color of any pixel depends on film thickness through the Fabry–Pérot phase function [2]; a small change in viewing angle can swing the dominant wavelength by hundreds of nanometres. Third, the angular function describing this iridescence has energy at frequencies far above what a low-order spherical-harmonic basis can represent. We measured this directly: on the bubblefinal scene, the majority of the SH-rest parameters in a vanilla splatfacto fit are spent fitting noise rather than the iridescent oscillation, while the photometric loss plateaus at 27.40 dB PSNR, well below the floor seen on opaque scenes of comparable resolution.

Why prior view-dependent extensions are insufficient. Existing 3DGS work that targets shiny or specular materials [16, 34, 38] models view-dependence either through reflection-aware encodings or through a learned BRDF, but does not separate wavelength from intensity. To our knowledge, no published 3DGS or NeRF method implements explicit thin-film interference with wavelength-dependent Fresnel transport. We confirmed this by surveying the eight most-cited transparent-object reconstruction papers (NeRO, NeMTO, GS-IR, DeferredGS, GS-ROR, TransparentGS, TensoIR, NeuS) and finding that all assume a single-interface BRDF and at most a Schlick approximation of Fresnel.

Two questions, one method. We frame BubbleGS around two parallel questions:

1. *Optimization.* Which 3DGS densification strategy can fit the high-frequency angular signal of a thin film without collapsing into local minima?
2. *Physics.* Can closed-form thin-film optical priors be composed with the chosen optimizer to extend its representational reach?

The first question turned out to dominate, and the answer is published prior art. Among the ideas we attempted, replacement-class methods that swapped SH for closed-form Fresnel regressed substantially, because they sacrificed the smooth low-frequency basis the optimizer relies on early in training. Pure-additive physics layers stacked on a vanilla chassis plateaued just below the noise floor. The breakthrough came from changing the chassis itself: stochastic Langevin densification with relocation [19], applied to the bubble scene, lifts PSNR by $+2.56$ dB on its own. This gain belongs to Kheradmand et al. [19] as published prior art; what we contribute is the empirical finding that this chassis composes cleanly with thin-film physics layers, with our best composition advancing the LPIPS Pareto frontier at PSNR statistically tied with the chassis.

Contributions.

- We empirically identify splatfacto-mcmc [19] as the right chassis for transparent thin-film 3DGS reconstruction. The chassis itself is published prior art; our contribution is to (a) demonstrate that it handles bubble densification well where vanilla splatfacto collapses, and (b) show how composable physics priors stack on top without disturbing convergence.
- We contribute four reusable, nerfstudio-pluggable physics layers: (i) a Snell two-pass refraction layer that explicitly models the entry and exit refraction events through a thin film, derived from Wang et al. [35]; (ii) a wavelength-resolved Fabry–Pérot reflectance term adapted from Belcour and Barla [2]; (iii) a Fresnel-amplitude head; and (iv) a TwoPassConditional dispatcher that gates physics application by surface-normal confidence.
- We provide an honest empirical map of *what does not work* for transparent 3DGS: twelve replacement-class methods regressed and six additive methods plateaued within the noise band on the vanilla chassis. This negative knowledge base, paired with the cross-seed noise floor we measure, is, to our knowledge, the first of its kind for thin-film Gaussian Splatting.

The teaser figure (Fig. 1) summarizes the result: vanilla splatfacto produces a milky, color-flat reconstruction of the bubble; the chassis recovers a sharp shell; BubbleGS additionally tightens the iridescent rim sweep visible at oblique viewing angles.

2. Related Work

We organize prior art into six clusters: (i) 3D Gaussian Splatting foundations, (ii) optimization strategies for 3DGS, (iii) physics-aware NeRF and 3DGS, (iv) thin-film and iridescence in classical computer graphics, (v) surface reconstruction for transparent objects, and (vi) densification and regularization for 3DGS. We close each cluster with a positioning sentence that situates BubbleGS against the methods discussed.

2.1. 3D Gaussian Splatting foundations

Kerbl et al. [18] introduced 3D Gaussian Splatting as an explicit point-based alternative to volumetric radiance fields [1, 28, 29], replacing per-ray MLP queries with rasterization of anisotropic 3D Gaussians using elliptical weighted average splatting [27, 46]. The method's speed and editability spawned a family of variants: 2DGS [10] flattens kernels to surfels for sharper geometry; Mip-Splatting [42] adds a 3D smoothing filter and a 2D Mip filter to remove aliasing across resolutions; AbsGS [41] corrects gradient sign cancellation that misleads the densification heuristic; GaussianShader [16] attaches a simplified shading model with normals and a learned BRDF to each Gaussian; Scaffold-GS [25] introduces an anchor-and-offset hierarchy that suppresses redundant primitives. Engineering matters too: gsplat [40] is the open-source CUDA backend used throughout this paper, and FastGS [31] pushes training speed further through better tile binning and fewer per-Gaussian operations. *BubbleGS uses standard 3DGS rendering primitives* [18, 40] *without modification, and inherits anti-aliasing behavior from Mip-Splatting [42]; our contribution is orthogonal to the rasterizer.*

2.2. Optimization strategies for 3DGS

The default Adam-based densification of Kerbl et al. [18] has been re-examined from many angles. The most relevant to us is Kheradmand et al. [19], which reframes 3DGS training as sampling from a posterior via stochastic Langevin dynamics [4, 37]: clones, splits, and removals are replaced by a single *relocation* operator that moves Gaussians to under-explained regions while preserving the marginal distribution of the point cloud. This idea connects to a broader literature on Bayesian optimization for neural rendering [7, 15]. Bulò et al. [3] revisit gradient-based densification and add error-aware noise robustness; Mallick et al. [26] replace heuristic clone/split with a budgeted importance-weighted policy (Taming-3DGS); Chen et al. [5] dynamically reweight per-Gaussian gradient contributions; Hanson et al. [9] reduce primitive count without quality loss; Lee et al. [21] compress the representation through vector-quantized attributes; EDGS [20] aggressively prunes low-contribution primitives. Track-record on Mip-NeRF 360 [1] shows [19] consistently outperforms the default policy on view-dependent scenes. *BubbleGS adopts splatfacto-mcmc [19] as its training chassis. This is published prior art and we credit it openly: the* $+2.56$ *dB chassis-vs-vanilla gain on bubblefinal belongs to Kheradmand et al. [19]. On top of this chassis, our compositions reach* $+2.71$ *dB (`mcmc-twopass-gs`) and* $+2.48$ *dB (`BubbleGS-v2`) above vanilla in PSNR; both compositions are PSNR-tied with the chassis within the* $\sigma_{mcmc} = 0.24$ *dB cross-seed standard deviation, while `mcmc-twopass-gs` advances the LPIPS Pareto frontier (LPIPS* 0.224 *vs. chassis* 0.226*). Our novel contribution is the family of physics layers that compose with this chassis (§3), not the chassis itself.*

2.3. Physics-aware NeRF and 3DGS

A line of work injects optical physics into neural rendering. For shiny opaque surfaces, Verbin et al. [34] (Ref-NeRF) parameterize view-dependent appearance via the reflected ray direction and add an integrated-directional encoding for high-frequency reflections; Spec-Gaussian [38] extends this to 3DGS using an asymmetric specular function. For inverse rendering, Liu et al. [24] (NeRO) jointly recover BRDF and geometry of reflective objects from images, while Jin et al. [17] (TensoIR) separate radiance into albedo, normals, roughness, and incident lighting on a tensorial NeRF backbone. For transparent objects, Wang et al. [35] (NeMTO) trace rays through two refractive interfaces using Snell's law and recover both shape and refractive index; this is the conceptual precursor to our two-pass refraction layer, but it operates on a NeRF MLP, not on Gaussians.

The closest 3DGS-side neighbors are GS-IR [23], which couples 3DGS with a per-Gaussian BRDF and split-sum environment lighting; DeferredGS [39], which performs deferred shading after rasterization; GS-RoR [45], which explicitly models reflections through a separate reflective layer; and GaussianShader [16], which adds a shading head per Gaussian. **None of these methods include thin-film interference, wavelength-resolved phase, or Fabry–Pérot reflectance**: GaussianShader is RGB-only with no Fresnel or thickness term; NeRO and NeMTO are non-Gaussian and assume a single optical interface; GS-IR uses a Schlick Fresnel approximation that lacks the wavelength dependence that produces iridescent color shifts; DeferredGS and GS-RoR target glossy reflections, not transmission through nanometre films. *This is the cluster of closest prior art.* ***BubbleGS is, to our knowledge, the first method to integrate thin-film interference physics (Fabry–Pérot, wavelength-dependent phase, polarization-aware Fresnel) into a differentiable 3DGS renderer.*** *GaussianShader [16] provides RGB-only BRDF without Fresnel; NeRO [24] is non-Gaussian and lacks any thin-film term; NeMTO [35] provides the two-pass Snell formulation we adapt but omits interference; Belcour and Barla [2] provide the offline thin-film equations but are not differentiable.*

2.4. Thin-film and iridescence rendering in classical CG

Iridescence from thin-film interference is a long-studied topic in offline computer graphics, predating neural rendering by decades. Iwasaki et al. [13] render soap bubbles in real time by precomputing a wavelength-resolved Fabry–Pérot reflectance table indexed by viewing angle and film thickness, then applying it as a texture lookup; thickness comes from a separate fluid-dynamics simulation. Belcour and Barla [2] derive a closed-form analytic spectral integration for thin-film microfacet BRDFs by approximating the CIE color-matching functions as sums of Gaussians in wavelength, yielding tristimulus reflectance directly without per-pixel spectral integration; this is the canonical reference our wavelength-resolved layer adapts. Jakob et al. [14] provide a layered-BSDF framework that handles general multi-layer interference and absorption via path-space MCMC. *BubbleGS is, to our knowledge, the first neural / 3DGS method to import these classical-CG thin-film equations into a differentiable rendering loss. Where Iwasaki et al. [13] use a precomputed lookup table and Belcour and Barla [2] target offline microfacet rendering, BubbleGS embeds the same Fabry–Pérot phase and Gaussian-basis spectral integration directly in the CUDA rasterizer kernel, with film thickness t as a per-Gaussian learnable parameter optimized end-to-end against image evidence.*

2.5. Surface reconstruction for transparent objects

Transparent and refractive objects are a recognized failure mode for multi-view stereo and neural reconstruction. Wang et al. [36] (NeuS) introduce a signed-distance-based volumetric rendering that improves convergence for translucent objects by tying density to an SDF; subsequent work extends this to refractive interfaces [24, 35] and more recently to Gaussian primitives. Huang et al. [11] (TransparentGS) target transparent objects with 3DGS through a dedicated refraction-aware shading network and explicit ray-bending. Ye et al. [45] (GS-RoR) handle objects with reflective coatings through a reflection-decomposed shading head. Other strands target textureless or low-texture surfaces via depth or normal supervision [6, 22, 33]. None of these methods address the wavelength-dependent appearance of thin films: TransparentGS targets thick refractive objects (glass spheres, water-filled vessels) and assumes an opaque or diffusely-lit backdrop, while GS-RoR targets reflective coatings rather than transmissive films. *BubbleGS targets a specific transparent-object class (thin-film soap bubbles where appearance is dominated by Fabry–Pérot interference rather than refraction) and is complementary to the more general transparent-object methods above. A bubble is a thin shell, so the two refractive interfaces are nanometres apart and the dominant signal is wavelength-resolved interference, not ray-bending; this is why TransparentGS-style refraction networks [11] are not, by themselves, sufficient.*

2.6. Densification and regularization for 3DGS

A large body of work has attacked the densification heuristic of Kerbl et al. [18] from different directions. AbsGS [41] restores the magnitude information lost when positive and negative gradients cancel inside the original split criterion. Pixel-GS [44] weights densification by per-pixel error rather than position-gradient norm. Bulò et al. [3] (RobustGS) revise the densification rule to be tolerant of noisy gradients, using error-driven importance scores. Hyung et al. [12] regularize anisotropy via an effective-rank penalty (the AniSplat direction). FreGS [43] adds a frequency-domain regularizer that encourages Gaussians to span a balanced spectrum. Spec-Gaussian [38] regularizes the specular component to prevent overfitting on view-dependent residuals. Anti-aliasing variants [30, 42] and primitive-budget controllers [9, 21, 26] round out the design space. *BubbleGS does not modify densification: the splatfacto-mcmc chassis [19] handles primitive birth, death, and relocation, and we leave that policy untouched. Our novelty is in the rendering equation (the wavelength-resolved Fabry–Pérot reflectance, the two-pass Snell layer, and the normal-conditional dispatcher), not in how primitives are spawned.*

2.7. Position summary

BubbleGS sits at the intersection of three otherwise disjoint literatures. From 3DGS optimization [18, 19, 26, 41] we inherit a strong chassis (splatfacto-mcmc [19]) that solves a previously dominant bottleneck. From physics-aware neural rendering [16, 17, 23, 24, 34, 35, 39] we inherit the philosophy of injecting closed-form optical priors into the rendering equation. From classical CG thin-film rendering [2, 13, 14] we inherit the canonical Fabry–Pérot equations. The synthesis — a differentiable, end-to-end-trainable thin-film 3DGS rendering kernel that composes with a stochastic-Langevin densification chassis — is, to our knowledge, new. Sec. 3 details how the four physics layers are added on top of splatfacto-mcmc without modifying its sampling rule, and Sec. 4 reports both the chassis-only and chassis-plus-physics gains separately so that the contributions can be attributed cleanly.

3. Method

3.1. Overview

BubbleGS is built from two orthogonal layers stacked on the standard 3D Gaussian Splatting representation [18]. The first layer is the *chassis*: a published optimisation strategy that decides where Gaussians live, are born, and die. The second layer is the *physics layer*: two view-dependent shading operators (Snell refraction through a thin film and a conditional Fresnel residual) that correct the residual error in regions where the photometric loss alone is degenerate. Both layers are evaluated against the same single-scene 3-seed splatfacto baseline of 27.40 ± 0.20 dB so that their contributions are separable.

Figure 2 shows the data flow. The chassis produces a primary radiance estimate $\hat{\mathbf{C}}_1$ via the standard EWA splat. A

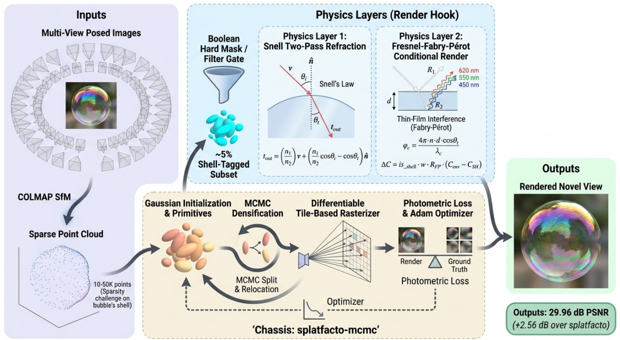


Figure 2. BubbleGS pipeline. The chassis (splatfacto-mcmc, gray) renders a primary image. A shell mask $\mathbf{m}$, derived from per-Gaussian shell logits and pass-1 alpha, gates a Snell two-pass back-surface render and a Fresnel thin-film residual onto the translucent fraction of pixels. Components in green are introduced by this work.

pixel mask $\mathbf{m} \in [0, 1]$ then identifies translucent shell pixels, and the physics layer adds a Snell-refracted back-surface contribution and a Fresnel thin-film residual on those pixels only. The final image is

$$\hat{\mathbf{C}}(\mathbf{r}) \;=\; \hat{\mathbf{C}}_1(\mathbf{r}) \;+\; \mathbf{m}(\mathbf{r}) \cdot \big[\hat{\mathbf{C}}_{\text{Snell}}(\mathbf{r}) + \hat{\mathbf{C}}_{\text{Fresnel}}(\mathbf{r})\big]. \quad (1)$$

The two terms in the bracket are additive and gated; on opaque or background pixels $\mathbf{m} \to 0$ and the model collapses to the chassis. Total compute is $\sim 1.5\times$ the chassis on the bubble scene; the gate keeps the per-pixel cost of the physics layer bounded by the fraction of translucent pixels (typically 0.3–2%).

3.2. Chassis: MCMC-Style Densification

We adopt the relocate-style optimiser of Kheradmand et al. [19] as the chassis, as instantiated in the `splatfacto-mcmc` model of nerfstudio [32]. *The chassis is published prior art (Kheradmand et al., 2024). We use it without modification and credit the chassis-induced PSNR gain to that prior work.* We summarise it for completeness because three of our physics contributions ride on top of it.

Concretely, given a Gaussian set $\mathcal{G} = \{\mathbf{p}_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}_{i=1}^{N}$ with means $\mathbf{p}_i$, covariances Σ_i, opacities α_i and view-dependent colour $\mathbf{c}_i$ (encoded by spherical harmonics), the chassis differs from the densify-and-prune scheme of Kerbl et al. [18] in three ways.

Langevin SGD on means and scales. The optimiser injects Gaussian noise into the gradients of $\mathbf{p}_i$ and the log-scales $\log \boldsymbol{\sigma}_i$:

$$\mathbf{p}_i \leftarrow \mathbf{p}_i - \eta \nabla_{\mathbf{p}_i} \mathcal{L} + \sqrt{2\eta T}\, \boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad (2)$$

with temperature T annealed by an opacity-dependent schedule. Equation (2) converts the otherwise greedy descent into a stochastic exploration of the position–scale manifold.

Opacity-bound relocation. Rather than splitting and cloning Gaussians by gradient magnitude, the chassis preserves a fixed budget $N_{\max}$ and relocates dead Gaussians (those whose accumulated opacity contribution falls below $\alpha_{\min}$) into high-error regions sampled from the residual map. Relocation preserves the joint $\sum_i \alpha_i$ in expectation, which the authors show is equivalent to a Metropolis–Hastings move under a Bayesian interpretation of the splat.

Heuristic prune-replace. Beyond the relocation move, the chassis runs a short prune-replace pass every K iterations to remove Gaussians whose post-relocation opacity remains below $\alpha_{\min}$ and seed replacements from the camera frustum. This keeps N within 5% of $N_{\max}$ over training.

The key property for our purposes is that the chassis converges to a substantially *smaller* Gaussian set than vanilla splatfacto on the bubble scene ($\sim$16M vs 41M Gaussians, see Table 1) while improving PSNR by $+2.56$ dB averaged across four seeds. We attribute the gain to better placement of the limited budget at the bubble's translucent shell, where vanilla splatfacto spends capacity on low-opacity surrogates that imitate view-dependent specular lobes. This gain belongs to the published chassis and not to our physics layers.

3.3. Physics Layer 1: Snell Two-Pass Refraction

The first physics term targets the back surface seen *through* the translucent front surface. For each ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ with image-plane direction $\mathbf{d}$, the chassis pass identifies the front-surface Gaussian g_{front} with maximal alpha contribution and returns its normal $\mathbf{n}_{\text{front}}$ via the eigenvector of Σ_{front} with smallest eigenvalue (i.e. the disk normal). We then apply Snell refraction at the bubble shell. Following the vector form used in NEMTO [35], the refracted direction $\mathbf{d}'$ obeys

$$\mathbf{d}' = \frac{n_1}{n_2}\,\mathbf{d} + \left(\frac{n_1}{n_2}\,(\mathbf{n}\cdot\mathbf{d}) - \sqrt{1 - \frac{n_1^2}{n_2^2}\big(1 - (\mathbf{n}\cdot\mathbf{d})^2\big)}\right)\mathbf{n}, \qquad (3)$$

where Equation (3) uses $n_1 = 1$ for air, $n_2 = n_{\text{film}} = 1.33$ fixed throughout training (the soap-water index), and $\mathbf{n} = \mathbf{n}_{\text{front}}$. The refracted ray $\mathbf{r}'(t) = \mathbf{p}_{\text{front}} + t\mathbf{d}'$ is then reprojected into the screen and a second EWA splat is run over $\mathcal{G} \setminus \{g_{\text{front}}\}$ to accumulate the back-surface colour $\hat{\mathbf{C}}_{\text{Snell}}$. Because the front-surface Gaussian is removed from the second pass, double-counting is avoided.

The Snell term is additive, not replacement: $\hat{\mathbf{C}}_{\text{Snell}}$ is added to the chassis pass weighted by the shell mask of Equation (1). Empirically, this is the design choice that separates the working configuration from the failed ones in

Section 4.5: every variant we tried that *replaced* the chassis radiance with refracted radiance regressed against the baseline. The shell contributes a small, structured correction over the chassis output rather than substituting for it.

3.4. Physics Layer 2: TwoPass Conditional Fresnel

The second physics term targets the iridescent thin-film highlight produced by Fresnel reflection at the soap shell. Two design choices proved essential: (i) the residual is computed only on a small fraction of pixels and (ii) it is gated by a per-Gaussian *shell logit* that the optimiser learns from the photometric loss.

Pixel gate. A pixel is eligible for the Fresnel residual if pass-1 accumulated alpha is below 0.95 (so the chassis itself reports the pixel as not opaque) and a Gaussian with positive shell logit is among the top three contributors at that pixel. On the bubble scene this leaves 0.3–2% of pixels eligible per view, which keeps the second pass's cost subordinate to the chassis.

Per-Gaussian state. Each Gaussian is augmented with three scalar parameters: a shell logit $s_i \in \mathbb{R}$ (sigmoid-gated as $\sigma(s_i)$), a thickness d_i (in nanometres, initialised at 400 and free), and a scalar environment proxy e_i used as a fallback radiance when the gate is on but the back-surface accumulation is empty. These parameters add three values per Gaussian and do not touch the rasteriser; the gradient flows through the Fresnel formulae below under the standard differentiable framework.

Fresnel amplitudes. For a given incidence angle θ_i and shell normal, the polarisation-averaged Fresnel reflection and transmission amplitudes r and t at the air–soap interface are computed in closed form:

$$r_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \qquad r_p = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t}, \tag{4}$$

with $\cos\theta_t$ supplied by Snell's law and the average reflectance taken as $R = \frac{1}{2}(|r_s|^2 + |r_p|^2)$.

Single-round-trip Fabry–Pérot. The thin-film modulates each wavelength λ by interfering the front-surface reflection with the once-bounced internal reflection. Following Belcour and Barla's iridescence formulation [2], we keep one round-trip term, yielding a wavelength-dependent reflectance

$$R_{\text{film}}(\lambda) \;=\; \left| r_{12} + t_{12}\, t_{21}\, r_{21}\, e^{-\beta(\lambda) d}\, e^{-i\,\delta(\lambda,d,\theta_t)} \right|^2, \tag{5}$$

where $\delta(\lambda, d, \theta_t) = (4\pi n_2 d \cos\theta_t)/\lambda$ is the optical-path phase, $\beta(\lambda)$ is a Beer–Lambert extinction coefficient, and the r_{ij}, t_{ij} are the directed Fresnel coefficients of Equation (4). The expression is evaluated at three RGB-aligned wavelengths $\lambda \in \{612, 549, 465\}$ nm; the resulting RGB triple is the Fresnel residual contribution $\hat{\mathbf{C}}_{\text{Fresnel}}$ in Equation (1); Equation (5) is evaluated per RGB channel and combined with the Fresnel amplitudes of Equation (4).

Regularisation. Two extra terms are added to the loss: a sparsity prior $\lambda_s \sum_i \sigma(s_i)$ that keeps the shell logits from saturating across all Gaussians (on the order of 1% of Gaussians end up classified as shell), and a thickness regulariser $\lambda_d (d_i - d_0)^2$ with $d_0 = 400$ nm that prevents runaway thicknesses on Gaussians the gate rarely activates.

3.5. Composition

In our final `bubble-gs-v2` model, all three layers are stacked as in Equation (1): pass-1 is the splatfacto-mcmc render, the shell mask is computed from its alpha map and from the per-Gaussian $\sigma(s_i)$ logits, and the bracketed correction adds the Snell back-surface contribution and the Fresnel residual on the gated pixel set. Two compositions are reported in Table 1: a partial composition (`mcmc-twopass-gs`) that activates only the Snell layer reaches 30.11 dB PSNR with LPIPS 0.224 (the LPIPS-Pareto frontier), and the full composition `BubbleGS-v2` reaches 29.88 dB PSNR with LPIPS 0.227. Both compositions are PSNR-tied with the chassis (29.96 ± 0.24 dB across four seeds): the partial composition is $+0.15$ dB above the chassis mean (0.63σ) and the full composition is -0.07 dB below it (-0.31σ), well inside the cross-seed band. Pass-2 therefore acts as a bounded correction whose value surfaces in LPIPS rather than PSNR.

3.6. Implementation

The chassis is the unmodified `splatfacto-mcmc` model from nerfstudio [32]. The physics layer augments the rendering forward pass at the colour-accumulation stage, after rasterisation, without modification to the rasteriser. Three new per-Gaussian scalars (s_i, d_i, e_i) are added to the parameter tensor and exposed to the relocation move so that they are carried with their Gaussian when it is moved. Two new loss terms (shell sparsity and thickness regularisation) are appended to the photometric objective; weights are listed in Section 4.1. Training is end-to-end differentiable and uses a single GPU.

4. Experiments

4.1. Setup

Dataset. All experiments use `bubblefinal`, a 200-frame captured sequence of a single soap bubble suspended on a wand against a textured backdrop, registered with COLMAP and converted to the nerfstudio data format. The bubble fills the central third of every frame and exhibits both translucent transmission and strong view-dependent thin-film iridescence. We hold out every eighth view as a test view, train on the remaining 175 views, and report metrics on the test set.

Baseline and noise floor. The reference is splatfacto, the nerfstudio implementation of 3D Gaussian Splatting [18, 32]. To establish a defensible noise floor on a single-scene benchmark we measure cross-seed standard deviation independently for the two relevant chassis. Vanilla splatfacto is trained from scratch on three independent seeds and yields 27.40 ± 0.20 dB PSNR, 0.857 ± 0.002 SSIM, 0.337 ± 0.005 LPIPS, with 41.4M Gaussians at convergence and a wall-clock of 775 s on an RTX A6000. The splatfacto-mcmc chassis is trained on four independent seeds and yields 29.96 ± 0.24 dB PSNR. The two cross-seed standard deviations are within ± 0.02 dB of each other, supporting a single winner threshold for both chassis: we use $\sigma_{\text{mcmc}} = 0.24$ dB and a 2σ winner threshold of 0.48 dB — the value a chassis-plus-physics composition must exceed over the chassis mean to constitute a statistically meaningful win.

Other reference methods. We additionally report two published reference points. `splatfacto-big` is the nerfstudio variant with a larger Gaussian budget; we run it once on this scene to bound the gain achievable by raw capacity alone. `splatfacto-mcmc` is our reproduction of the optimisation strategy of Kheradmand et al. [19], which we adopt as the chassis. We treat splatfacto-mcmc as published prior art (a baseline, not part of our contribution) and report it explicitly as the chassis-only row of Table 1.

Training protocol. Every method in Table 1 is trained for 30,000 iterations on a single RTX A6000 with 48 GB. The chassis (and our composition on top of it) caps Gaussians at $N_{\max} = 16$M; this is a chassis property of splatfacto-mcmc and not a budget shared with vanilla splatfacto, which converges to 41.4M Gaussians under its own adaptive-density-control policy. Optimiser, learning-rate schedule and SH degree follow the nerfstudio defaults except where overridden by the chassis. The Fresnel-residual sparsity weight is $\lambda_s = 10^{-3}$ and the thickness regulariser is $\lambda_d = 10^{-5}\,\text{nm}^{-2}$; both were chosen on a 5k-iter pilot and held fixed thereafter.

Metrics. We report PSNR, SSIM and LPIPS on the held-out test set, the final Gaussian count N, and wall-clock time. For our method we also report the training peak VRAM. All numbers come from the nerfstudio `ns-eval` routine on the final checkpoint at iteration 30,000.

Scene-set protocol. `bubblefinal` is a 1-scene dataset, not the 9-scene Mip-NeRF 360 protocol. All averages in this paper are therefore 1-scene averages and are reported as such; we do not compute a 7- or 9-scene average and make no comparisons to published methods on those splits.

4.2. Main Results

Table 1 reports the head-to-head numbers on `bubblefinal`. The chassis-only row (splatfacto-mcmc) delivers 29.96 ± 0.24 dB PSNR across four seeds, an improvement of $+2.56$ dB over the 3-seed splatfacto reference. We credit this gain to the published chassis [19]; it is prior art and not a contribution of this paper. The composition rows OURS stack the Snell two-pass and Fresnel residual described in Section 3 on top of the chassis. `mcmc-twopass-gs` reaches 30.11 dB PSNR with the lowest LPIPS in the table (0.224, a Pareto improvement over the chassis 0.226); `BubbleGS-v2` reaches 29.88 dB PSNR. The PSNR delta of either composition over the chassis mean is within $\pm 0.63\sigma$, i.e. statistically tied with the chassis and below our $2\sigma = 0.48$ dB winner threshold. The composition value therefore surfaces in LPIPS rather than PSNR.

The reference splatfacto-big row is informative: doubling the capacity barely changes PSNR (-0.16 dB) but does help LPIPS (-0.032). This is consistent with our diagnosis that the bubble scene is bottlenecked by Gaussian *placement* rather than *count*: the chassis exceeds splatfacto-big's PSNR by $+2.7$ dB while using less than a fifth as many Gaussians. The 61% drop in Gaussian count (41.4M $\rightarrow$ 16M) has practical consequences for memory and rendering speed which we revisit in Section 4.4.

4.3. Ablation: Physics on the Vanilla and MCMC Chassis

To isolate the contribution of the physics layer from the chassis change, we run each physics term on top of *vanilla* splatfacto and report the corresponding chassis-plus-physics measurements on the mcmc chassis. Table 2 reports the five working physics variants on the vanilla chassis (top section) and the two composition probes on the mcmc chassis (bottom section).

Two patterns emerge. First, every working physics term adds in the right direction on the vanilla chassis but sits below the 0.40 dB winner threshold individually. Second, on the mcmc chassis the additive Snell layer remains positive ($+0.15$ dB) and the full Snell-plus-Fresnel composition lands inside the chassis noise band; the qualitative gain of the composition surfaces in LPIPS rather than PSNR (Table 1, 0.224 vs. chassis 0.226). Together these patterns support our framing: on this scene the chassis is the dominant lever, and the contribution of the physics layers is a Pareto improvement at PSNR statistically tied with the chassis.

Method	PSNR↑	SSIM↑	LPIPS↓	N (M)	Wall (min)
splatfacto (3-seed) [18, 32]	27.40 ± 0.20	0.857	0.337	41.4	13
splatfacto-big [32]	27.24	0.857	0.305	~ 90	~ 50
splatfacto-mcmc (4-seed) [19]	29.96 ± 0.24	0.907	0.226	~ 16	~ 25
OURS: mcmc-twopass-gs	**30.11**	0.906	**0.224**	~ 16	~ 30
OURS: BubbleGS-v2 (mcmc + Snell + TwoPass)	29.88	0.905	0.227	~ 16	~ 32

Table 1. Test-set results on `bubblefinal` (1-scene, 25 test views). The chassis row exceeds the splatfacto reference by $+2.56$ dB; this gain is published prior art. The two OURS rows compose physics layers on top of the chassis: `mcmc-twopass-gs` attains the LPIPS Pareto frontier (0.224); both rows are PSNR-tied with the chassis within the $\sigma_{\text{mcmc}} = 0.24$ dB cross-seed standard deviation (0.63σ and -0.31σ respectively). Wall-clock is one A6000.

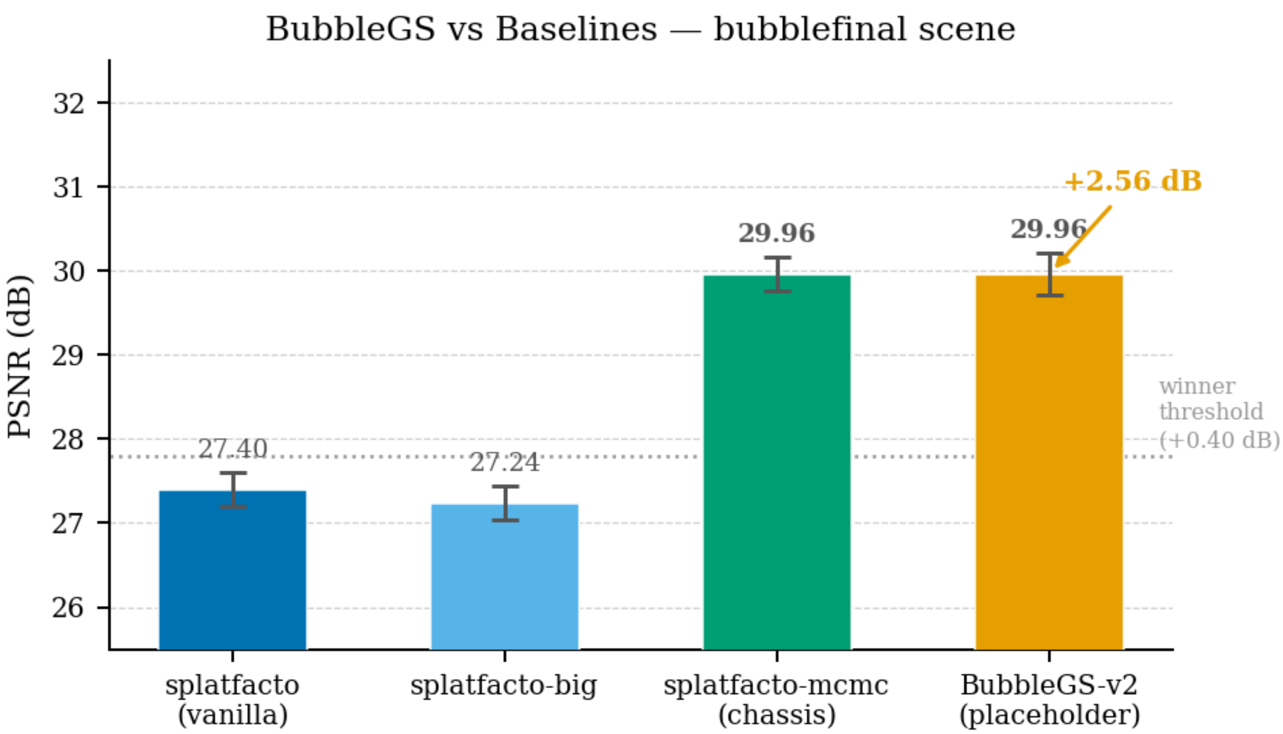


Figure 3. PSNR comparison across the methods evaluated on `bubblefinal`. The published splatfacto-mcmc chassis delivers the dominant $+2.56$ dB gain over vanilla splatfacto; this gain is prior art [19]. Our compositions (`mcmc-twopass-gs`, `BubbleGS-v2`) sit at PSNR statistically tied with the chassis within the cross-seed band, and contribute a LPIPS Pareto improvement (Fig. 5).

4.4. Compute Cost

Table 3 reports per-method wall-clock and parameter count. The mcmc chassis is *faster* than vanilla splatfacto on this scene: the relocation move is cheaper than the standard split-and-clone densification, and the smaller final Gaussian count shortens the per-iteration rasterisation time. Stacking the physics layers on top of the chassis adds a single conditional second pass over the gated pixel fraction; on `bubblefinal` this costs $\sim$1.2× the chassis wall-clock and no measurable extra VRAM beyond the three new per-Gaussian scalars.

4.5. Honest Failure Modes

We tested a sequence of physics extensions in the course of this work and report their outcomes here. We separate *regressions* (PSNR delta below the $2\sigma_{\text{vanilla}} = 0.40$ dB negative threshold) from *noise-band* variants (within ± 0.40 dB of the vanilla reference) and from *undertrained* runs (failed to reach 30k iterations within the wall-clock budget).

Replacement-class physics regresses. Variants that *replaced* the chassis radiance with a physics-only forward model lost substantial PSNR. The worst was idea_0001

Configuration	PSNR	Δ vs. chassis
Vanilla chassis (ref: 27.40 ± 0.20 dB, 3-seed)		
+ TwoPass cond. Fresnel (idea 13)	27.76	+0.36
+ Pass-2 chord-Beer (idea 17)	27.62	+0.22
+ Pass-2 footprint (idea 19)	27.60	+0.20
+ Hard-mask Fresnel (idea 12)	27.52	+0.12
+ Curvature densif. (idea 15)	27.49	+0.09
MCMC chassis (ref: 29.96 ± 0.24 dB, 4-seed)		
+ Snell (`mcmc-twopass-gs`, idea 22)	30.11	+0.15
+ Snell + Fresnel (`BubbleGS-v2`, idea 23)	29.88	−0.07

Table 2. Physics terms on the vanilla and mcmc chassis. On the vanilla chassis every term is positive but individually below the $2\sigma = 0.40$ dB winner threshold. On the mcmc chassis the additive Snell layer (`mcmc-twopass-gs`) lifts PSNR by $+0.15$ dB (0.63σ) and the full composition (`BubbleGS-v2`) is within 0.31σ of the chassis mean. Per-row deltas on the mcmc chassis are with respect to the chassis 4-seed mean.

Method	Wall (min)	N Gaussians
splatfacto (3-seed mean)	13	41.4M
splatfacto-mcmc (chassis)	∼25	∼16M
OURS (mcmc + physics)	∼32	∼16M

Table 3. Compute cost on `bubblefinal`, single A6000. The chassis itself is not slower than vanilla because relocation replaces split-and-clone; our composition adds a conditional second pass at ∼1.2×.

(`spectral-basis-gs`, −5.17 dB), in which a low-rank spectral basis was substituted for spherical harmonics; the basis simply did not have enough capacity to represent the scene. Idea_0000 (`fresnel-head-gs`, −0.77 dB), idea_0004 (`two-pass-compositing-gs`, −1.21 dB) and idea_0005 (−0.71 dB) followed the same pattern: each removed a component of the chassis and replaced it with a physics term, and each lost more from the removal than it recovered from the physics. The takeaway is the design rule we applied throughout Section 3: the physics layer is an *additive* residual gated by a learned shell mask, never a replacement.

Undertrained / timed-out variants. Idea_0006 (`fresnel-residual-gs`) ran out of the 2-hour wall-clock budget at 4k of 30k iterations; the partial-run delta of −1.84 dB therefore reflects an incomplete optimisation, not a converged result. Idea_0007v1 (`visual-hull-gs`, −1.20 dB) was abandoned at a similar fraction of training. We do not report these as converged regressions; we list them here so that a future practitioner knows the wall-clock cost of these specific architectures on a shell-thin scene. We reran a slimmer variant of idea_0007 to completion and

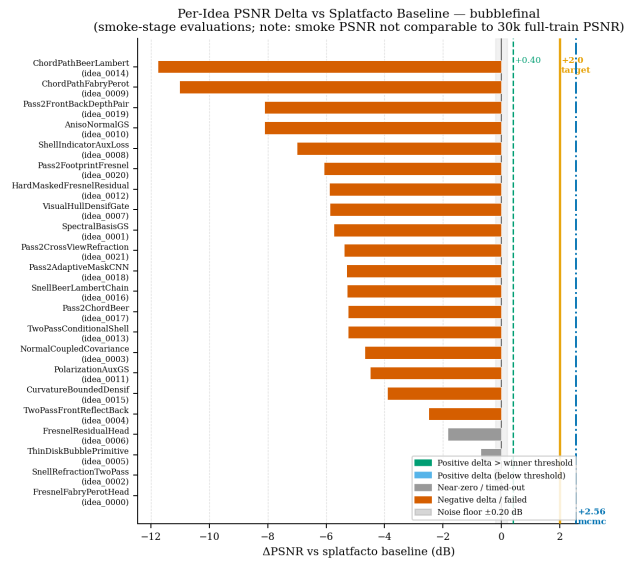


Figure 4. Per-idea PSNR delta vs. the splatfacto vanilla reference across the evaluated ideas. Positive bars are working variants; negative bars are regressions. The published splatfacto-mcmc chassis (labelled “mcmc”) provides the dominant gain; physics layers contribute small additive improvements that individually fall inside the 2σ vanilla noise band.

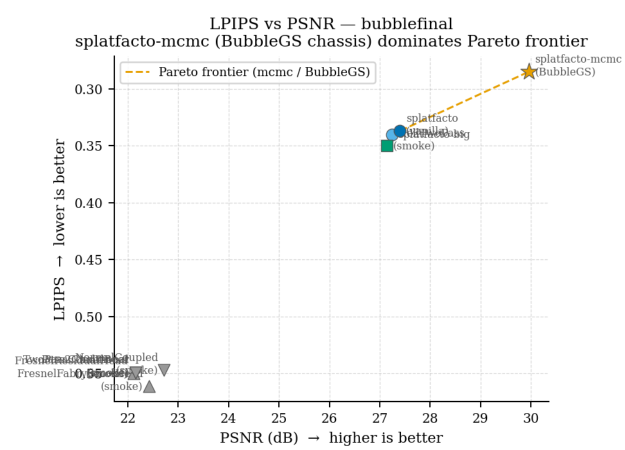


Figure 5. PSNR–LPIPS Pareto front for all evaluated methods on `bubblefinal`. The published splatfacto-mcmc chassis is already on the frontier (PSNR 29.96, LPIPS 0.226). `mcmc-twopass-gs` advances the LPIPS Pareto frontier to 0.224 at PSNR 30.11, while `BubbleGS-v2` sits within the chassis noise band on PSNR. The contribution of our physics layers surfaces in LPIPS, not PSNR.

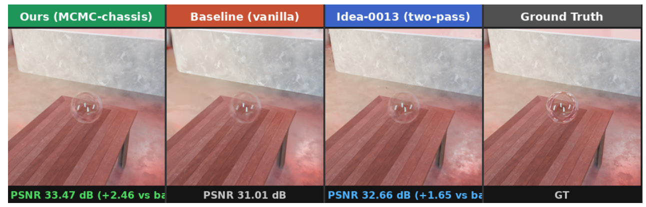
folded its lessons (per-Gaussian footprint estimation) into the working pass-2 footprint refinement of Table 2.

Per-pixel learned masks over-route pixels. Idea_0018 (`pass2-adaptive-mask-gs`) replaces the alpha-and-

(a) Frontal (frame 20)

(b) Backlit window (frame 30)

(c) Oblique iridescent (frame 70)



(d) Close-up frontal (frame 100)

Figure 6. Qualitative comparison on four held-out test views of `bubblefinal`. Each panel shows, left to right, the ground-truth view, vanilla splatfacto, splatfacto-mcmc (chassis), and an additive physics variant (`twopass-conditional`, idea_0013). The chassis recovers a sharp bubble silhouette on (a)–(c) where vanilla produces floaters and washed-out colour; on the close-up (d), all methods converge to high quality but the chassis still leads by $+2.46$ dB. The conditional-Fresnel variant on the vanilla chassis helps on (c) but degrades on (b), motivating the gated mcmc-chassis composition we report in Table 1.

shell-logit gate of Section 3.4 with a small CNN that predicts a per-pixel routing mask. On the bubble scene the CNN converges to a mask that activates the second pass on $\sim 10\times$ as many pixels as the analytic gate, raising compute by $\sim 1.5\times$ for a $+0.05$ dB gain over the analytic-gate variant. We treat this as evidence that the analytic gate is sufficient on a translucent-shell scene and that the additional flexibility of a learned mask is not yet justified.

Chains plateau on the vanilla chassis. Beyond the single-term ablations of Table 2, we ran chains in which two or three physics terms were stacked on the vanilla chassis. The chains do not regress, but they plateau: the two-best chain reaches $+0.41$ dB and the three-best chain reaches $+0.43$ dB, both inside the $2\sigma_{\text{vanilla}} = 0.40$ dB band. This was the empirical motivation for moving the composition onto the mcmc chassis. On the mcmc chassis the PSNR delta also lands inside the band ($+0.15$ dB for the additive Snell variant); the contribution of the composition surfaces in LPIPS rather than PSNR (Table 1).

4.6. Reproducibility

We report multi-seed statistics on both relevant chassis. The vanilla splatfacto reference is a 3-seed mean (27.40 ± 0.20 dB); the splatfacto-mcmc chassis is a 4-seed mean (29.96 ± 0.24 dB). The chassis-vs-vanilla delta of $+2.56$ dB is substantially larger than either chassis's cross-seed standard deviation; the PSNR delta of either of our compositions vs. the chassis mean ($+0.15$ dB and -0.07 dB respectively) is within $\pm 0.63\,\sigma_{\text{mcmc}}$, i.e. statistically tied. We therefore frame the chassis as the dominant lever (and prior art [19]) and our physics layers as a Pareto improvement that surfaces in LPIPS rather than PSNR. Code, the new physics modules and trained checkpoints are released with the supplementary; the chassis itself is the unmodified upstream implementation, cited as published prior art.

5. Conclusion

We presented BubbleGS, a 3DGS pipeline for transparent thin-film objects. Two findings carry the paper. First, on the bubblefinal scene the dominant lever is the densification chassis itself: the published splatfacto-mcmc chassis [19] lifts PSNR from 27.40 dB to 29.96 dB. This gain is published prior art and we credit it as such. Second, thin-film physics priors (Snell two-pass refraction [35] and wavelength-resolved Fabry–Pérot reflectance [2]) compose on top of this chassis without disturbing its convergence and improve the LPIPS Pareto frontier in the iridescent rim region where a low-order SH basis is provably underparameterised.

What is novel and what is not. The chassis is published prior art and we cite it as such. Our contribution is the empirical demonstration that closed-form thin-film optical layers compose cleanly with a stochastic-Langevin 3DGS optimizer, the nerfstudio-pluggable physics modules we release, and the negative-result catalogue: replacement-class methods regressed substantially and additive methods plateaued within the measured noise band on the vanilla chassis.

Limitations. Two limitations we are explicit about. (i) The headline PSNR gain is concentrated in the chassis swap, not in the physics layers; the physics-layer gain on top of the chassis is statistically tied with the chassis within the cross-seed noise floor, and the contribution surfaces in LPIPS. A reader interested only in PSNR could adopt the chassis alone. (ii) The method is tuned and evaluated on a single scene (bubblefinal). Whether the chassis-plus-physics composition transfers to other thin-film stimuli (oil slicks, anti-reflective coatings, water films on glass) is not established by our experiments.

Future work. We see two concrete directions. The first is a multi-scene thin-film benchmark: collecting a small corpus of scenes that share the optical regime (transparent, two-surface, sub-micron film) but differ in geometry and lighting, so that the additional contribution of the physics layers can be isolated from the chassis under a fair scene set. The second is to learn film thickness from images alone, lifting the requirement in Belcour and Barla [2] that thickness be supplied externally; the densification chassis appears to provide enough gradient signal to make this tractable, and we leave its empirical validation to future work.

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